

Github Link: <https://github.com/arin33/blood-donation-management>

Project Idea: **Blood Management System**

Project Overview:

The Blood Donation Management System is a software application designed to manage blood donor information efficiently. It is aimed at helping organizations or foundations like the **Amrai Agamir Blood Donation Foundation** track and manage blood donations. This system allows users to add, update, delete, search, and count blood donors based on their blood groups, along with other essential details such as contact information and city.

Key Features this project:

1. **Add Donor:** Users can add a new donor to the system by providing details such as name, blood group, contact information, and city.
2. **Search Donor by Blood Group:** The system allows users to search for donors based on their blood group. It will display all donors who match the given blood group.
3. **Display All Donors:** Users can view the complete list of all registered donors, along with their details.
4. **Delete Donor:** If a donor wishes to be removed or their details are no longer required, the system provides an option to delete a donor based on their name.
5. **Update Donor Information:** The system allows for updating donor details like blood group, contact information, or city. Users can search for a donor by name and modify their information.
6. **Count Donors by Blood Group:** The system provides a feature to count the number of donors for a specific blood group, which can help in determining the availability of blood types for donation.
7. **Exit:** The system will exit the application when the user chooses the "Exit" option from the menu.

Technologies Used Blood Management system:

- **Programming Language:** Java
- **Data Structure:** ArrayList (for storing donor information)
- **Scanner:** For taking user input via console.

Workflow in this Project:

1. **User Interface:** The system presents a simple text-based menu to the user, where they can choose one of the options:
 - Add a donor
 - Search for a donor by blood group
 - Display all donors
 - Delete a donor
 - Update donor information
 - Count donors by blood group
 - Exit the system
2. **Operations:**
 - When a user selects an option, the system performs the corresponding action, such as adding, searching, updating, or deleting a donor from the list.
 - For searching, the system checks each donor's blood group to find matches.

- When updating or deleting a donor, the system searches for the donor by name and modifies or removes the corresponding record.
- The count of donors for a particular blood group is calculated by iterating through the list of donors.

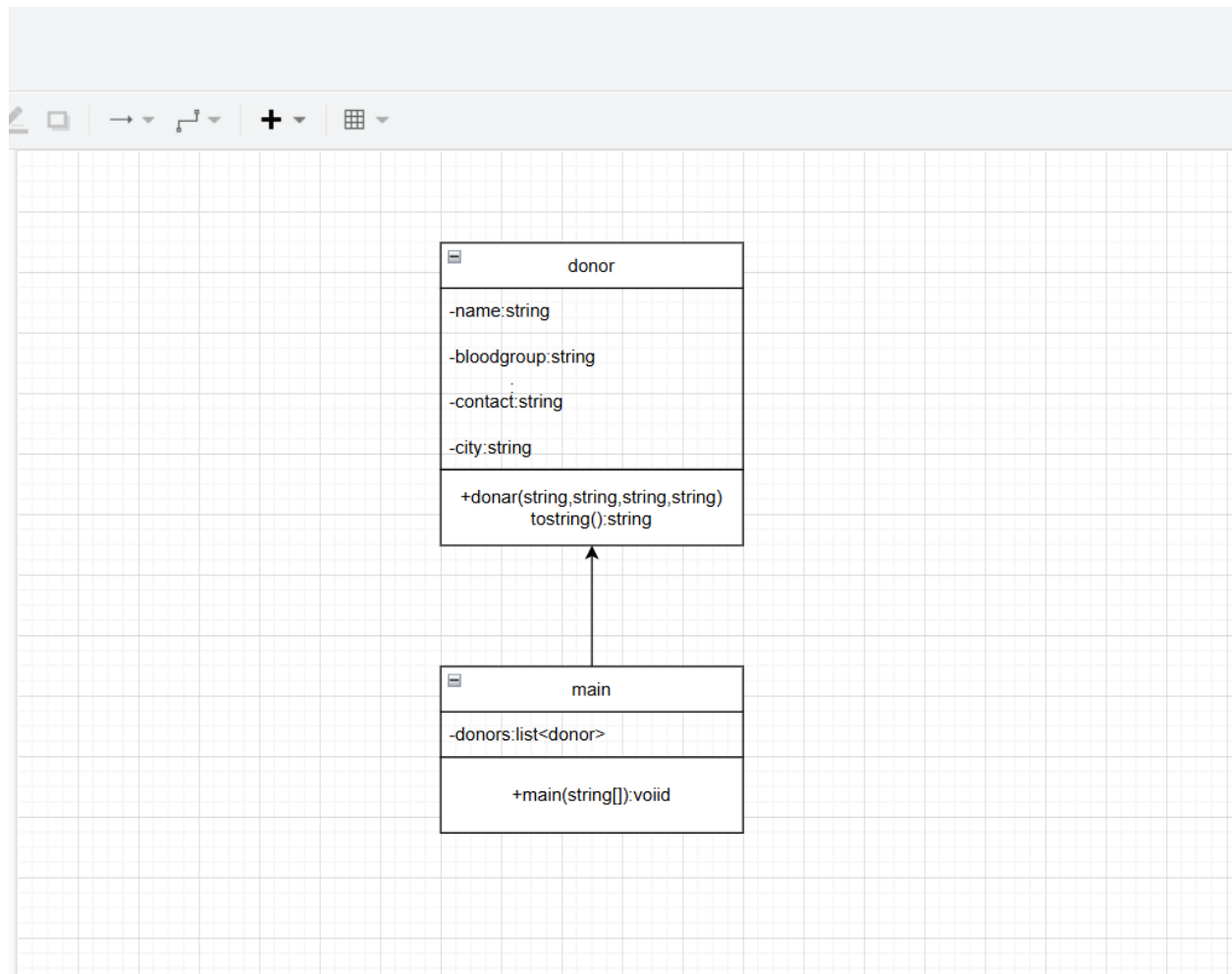
3. **Data Handling:**

- All donor information is stored in an ArrayList of Donor objects. Each Donor object holds details like the donor's name, blood group, contact number, and city.

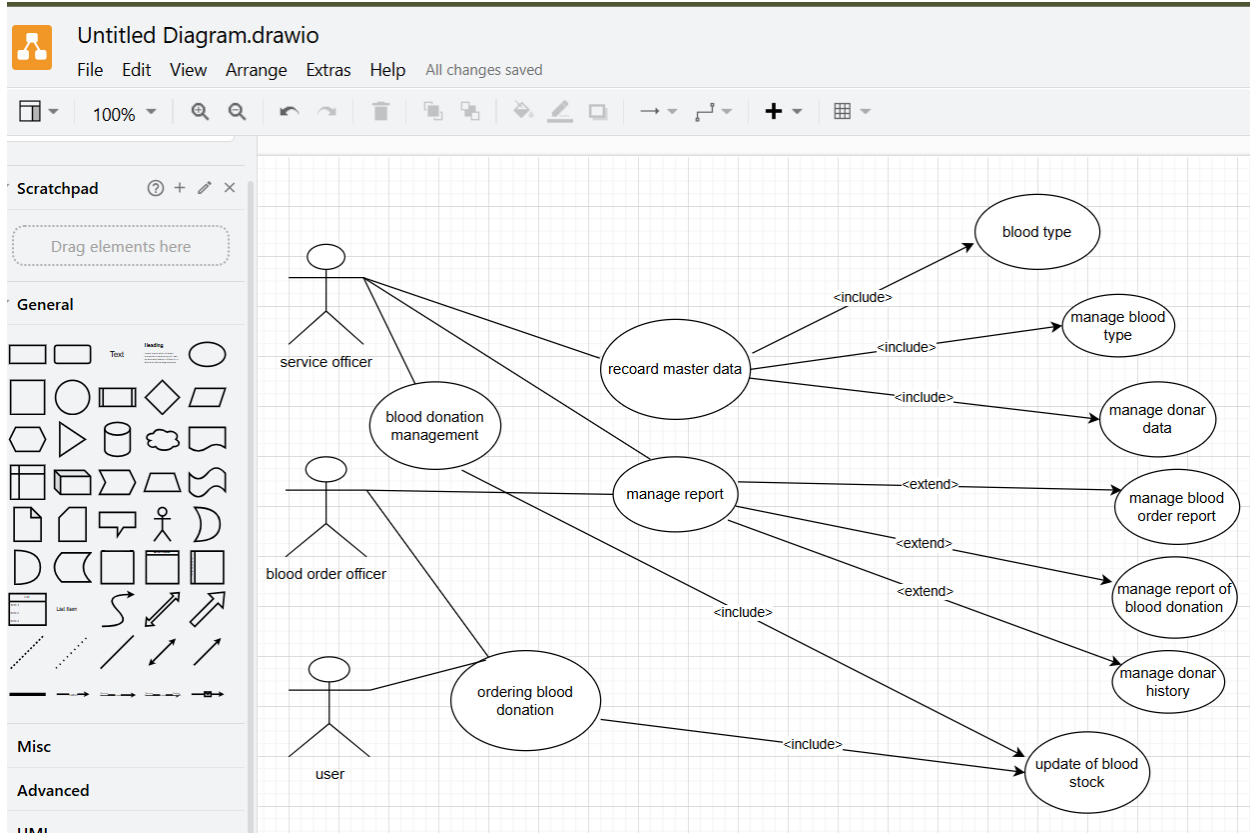
4. **Termination:** The application runs in a loop until the user chooses to exit.

This project is a great way to practice Java programming and learn how to manage data using collections, as well as build a simple console-based application for real-world applications like blood donation management.

Class diagram:



Use case diagram:



Activity diagram:

